Developer Technology Trends: What Developers Use Today and Want Next

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OUTLINE



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EXECUTIVE SUMMARY



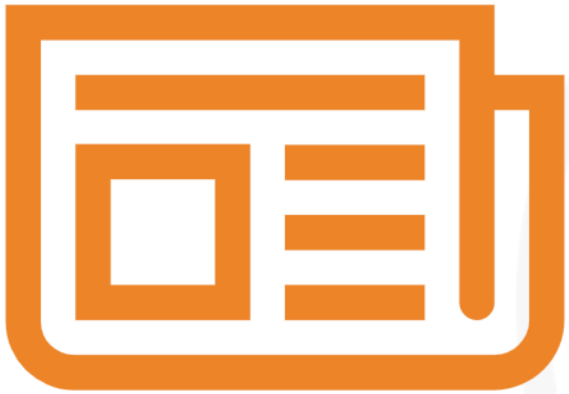
- **Purpose:** Analyzed Stack Overflow Developer Survey data to understand which technologies developers use today and want to use next
- **Approach:**
  - Data collection and wrangling: cleaned the survey data and handled missing values and duplicates
  - Analysis: Queried and explored the data using SQL and Python in Jupyter Notebooks
- **Visualization:** Built an interactive Cognos Analytics dashboard covering current usage, future trends and demographics
- **Key findings:** JavaScript (14,943) and SQL (12,602) lead languages. PostgreSQL leads databases (11,514), and AWS leads platforms
- **Future trends:** Demand stays strong for JavaScript, SQL, HTML/CSS and TypeScript. PostgreSQL grows to 12,193 as the most wanted database and Rust and Go enter the top 10 wanted languages
- **Recommendations:** Prioritize JavaScript/TypeScript, SQL, Python, PostgreSQL and AWS. Watch Rust, Go, Supabase and FastAPI

INTRODUCTION



- **Background:** Technology changes fast and developers and organizations need to know which tools are in use and which are gaining interest
- **Data source:** The Stack Overflow Developer Survey, with responses from developers around the world
- **Objective:** Identify current and future technology trends and understand who the respondents are
- **Key Questions:**
  - Which languages, databases, platforms and web frameworks do developers use today?
  - Which ones do they want to use next and what is the demographic profile of respondents (age, education, country)?

METHODOLOGY



- **Data source:** Stack Overflow Developer Survey dataset
- **Data preparation:** Cleaned the data using Python (Pandas) in Jupyter Notebooks, removing duplicates and handling missing values
- **Analysis:** Used SQL and Python in Jupyter Notebooks to count how many respondents use or want to use each technology
- **Visualization:** Built an interactive Cognos Analytics dashboard with three tabs
  - Current and future technology: Top 10 languages, databases, platforms and web frameworks (worked with vs. want to work with)
  - Demographics: respondent age, education level and country

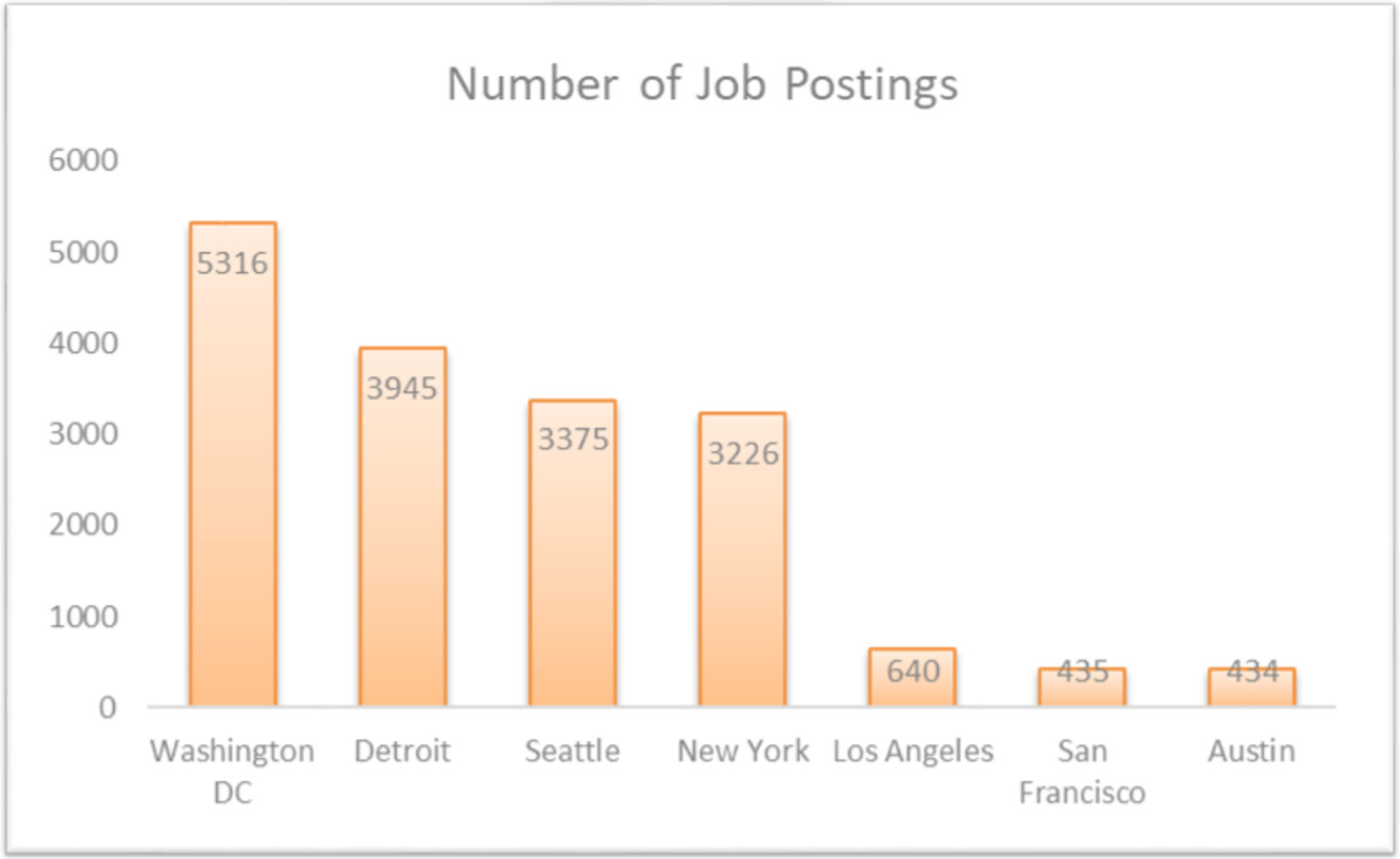
RESULTS

- **Current usage:** JavaScript (14,943), SQL (12,602) and HTML/CSS (12,410) are the most used languages
  - PostgreSQL (11,514) leads databases and AWS leads platforms
  - Node.js and React lead web frameworks

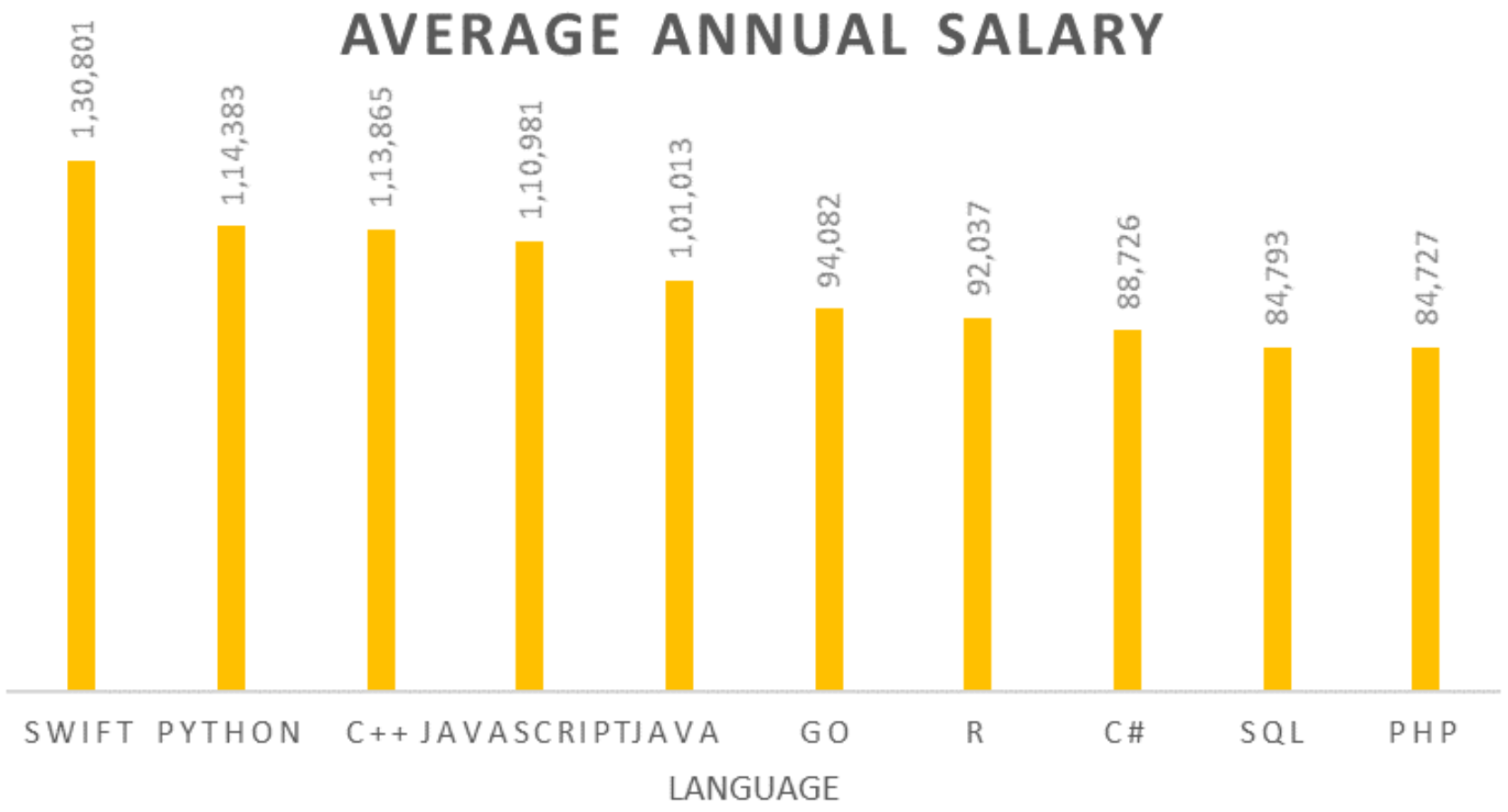
- **Future trends:** JavaScript, SQL, HTML/CSS and TypeScript remain the most wanted languages
  - PostgreSQL rises to 12,193 as the most wanted database
  - Rust and Go enter the top 10 wanted languages

- **Demographics:** Respondents are mostly aged 25-34 (41.3%) and 35-44 (27.3%)
  - A Bachelor's degree is the most common education level (8,629)
  - The United States and India have the most respondents

JOB POSTINGS



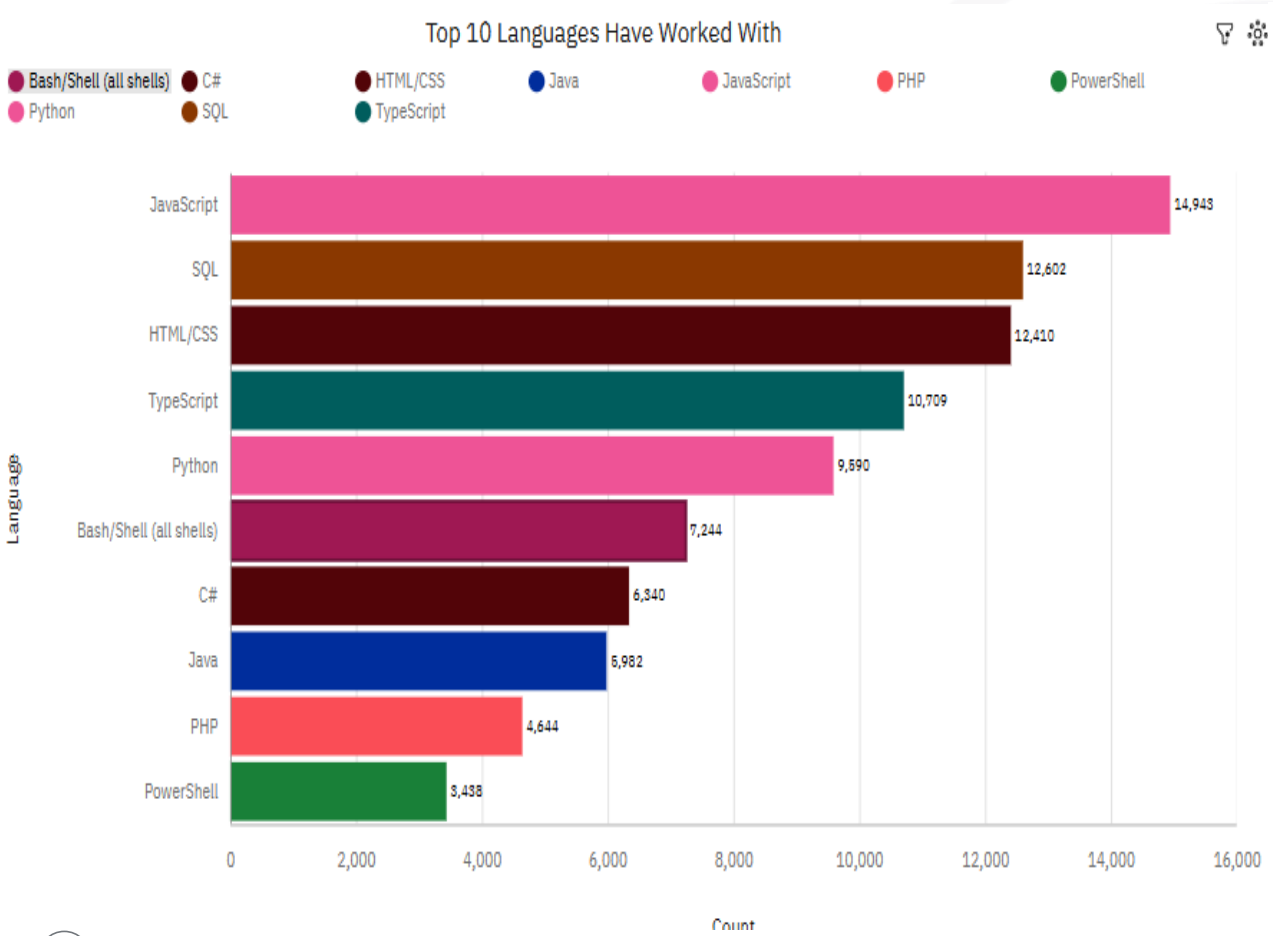
POPULAR LANGUAGES



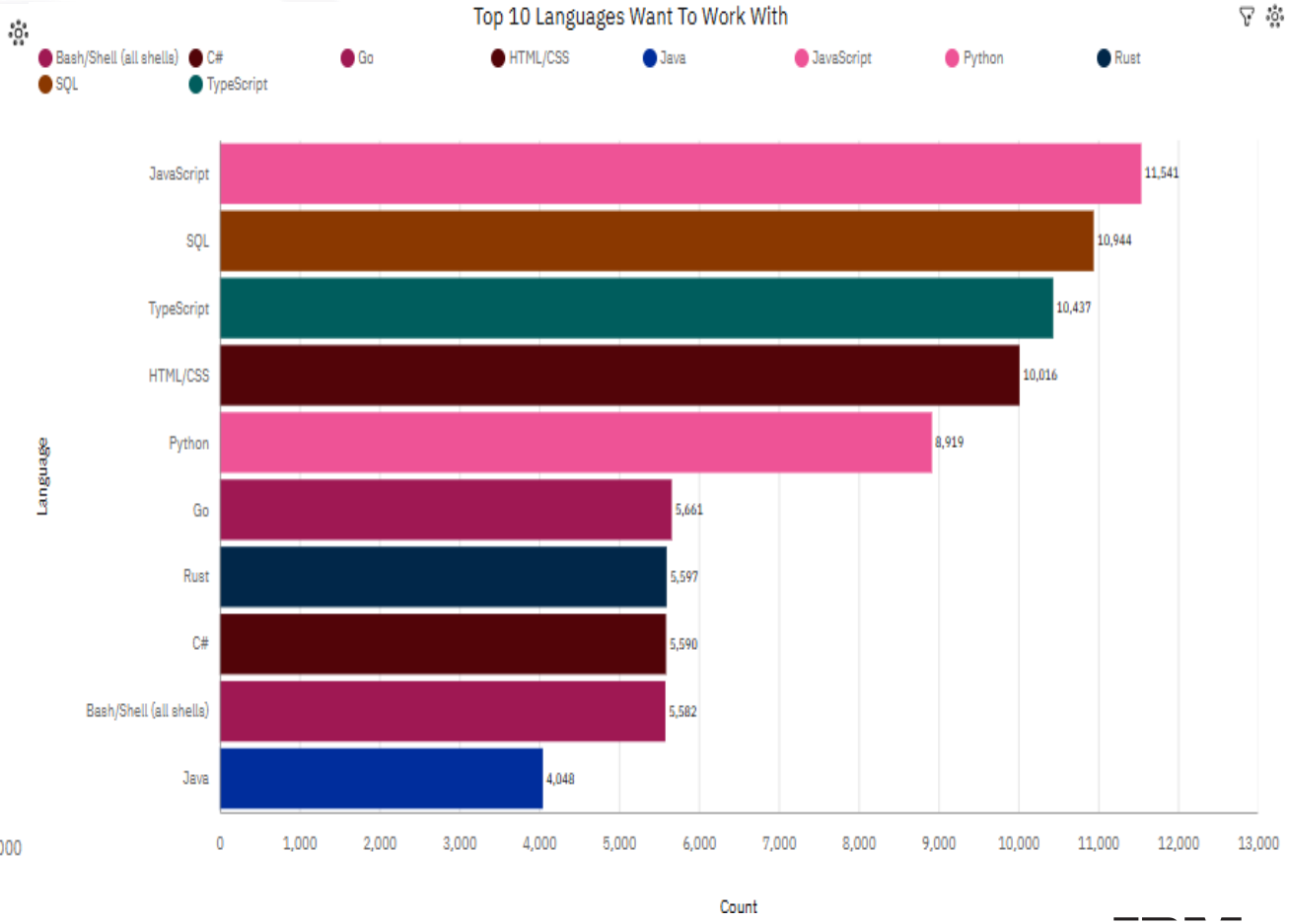
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PROGRAMMING LANGUAGE TRENDS

Current Year



Next Year



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

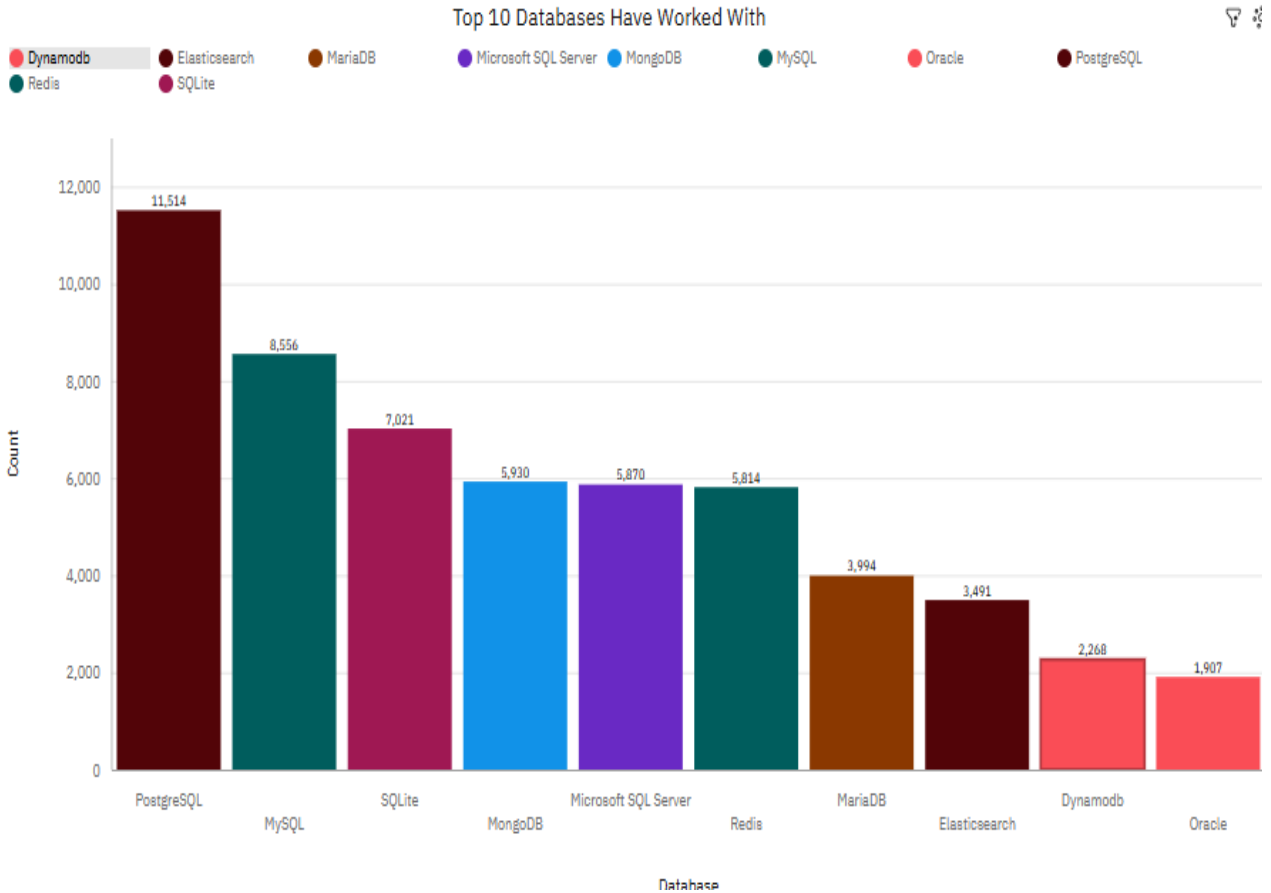
- JavaScript (14,943), SQL (12,602), and HTML/CSS (12,410) are the most-used languages, and JavaScript is still the most wanted (11,541)
- TypeScript (10,709 to 10,437) and Python (9,590 to 8,919) keep most of their interest
- Rust (5,597) and Go (5,661) enter the top 10 wanted, while Java falls from 5,982 to 4,048

Implications

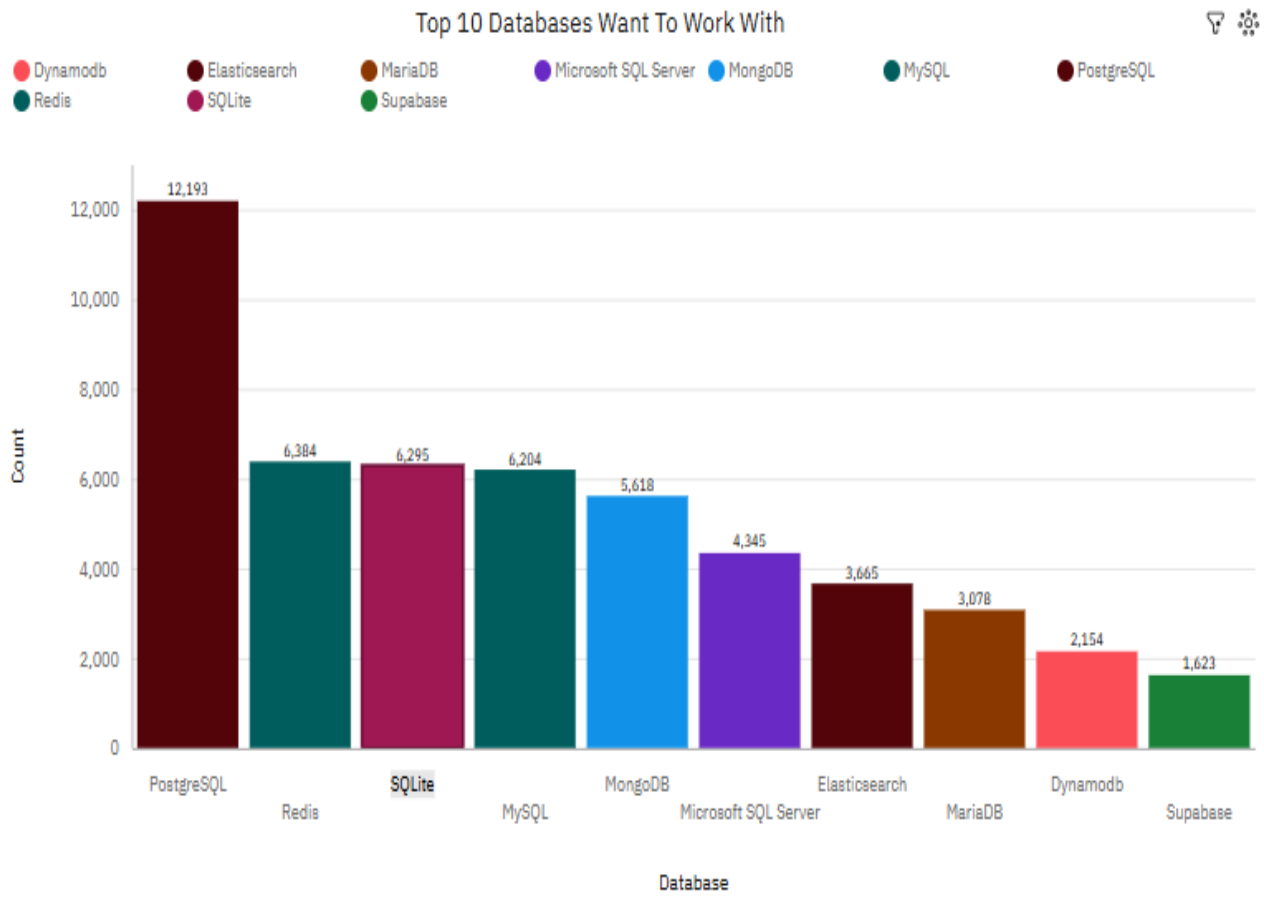
- JavaScript, SQL, and HTML/CSS remain the safest core skills to build on
- TypeScript and Python are strong additions to that core
- Rust and Go are worth watching as emerging languages, and Java's lower interest suggests demand may be cooling

DATABASE TRENDS

Current Year



Next Year



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- PostgreSQL is both the most-used (11,514) and the most-wanted (12,193) database
- MySQL is the second most used (8,556), but its interest falls to 6,204
- Redis rises from 5,814 to 6,384, and Supabase enters the wanted list (1,623) while Oracle drops out

Implications

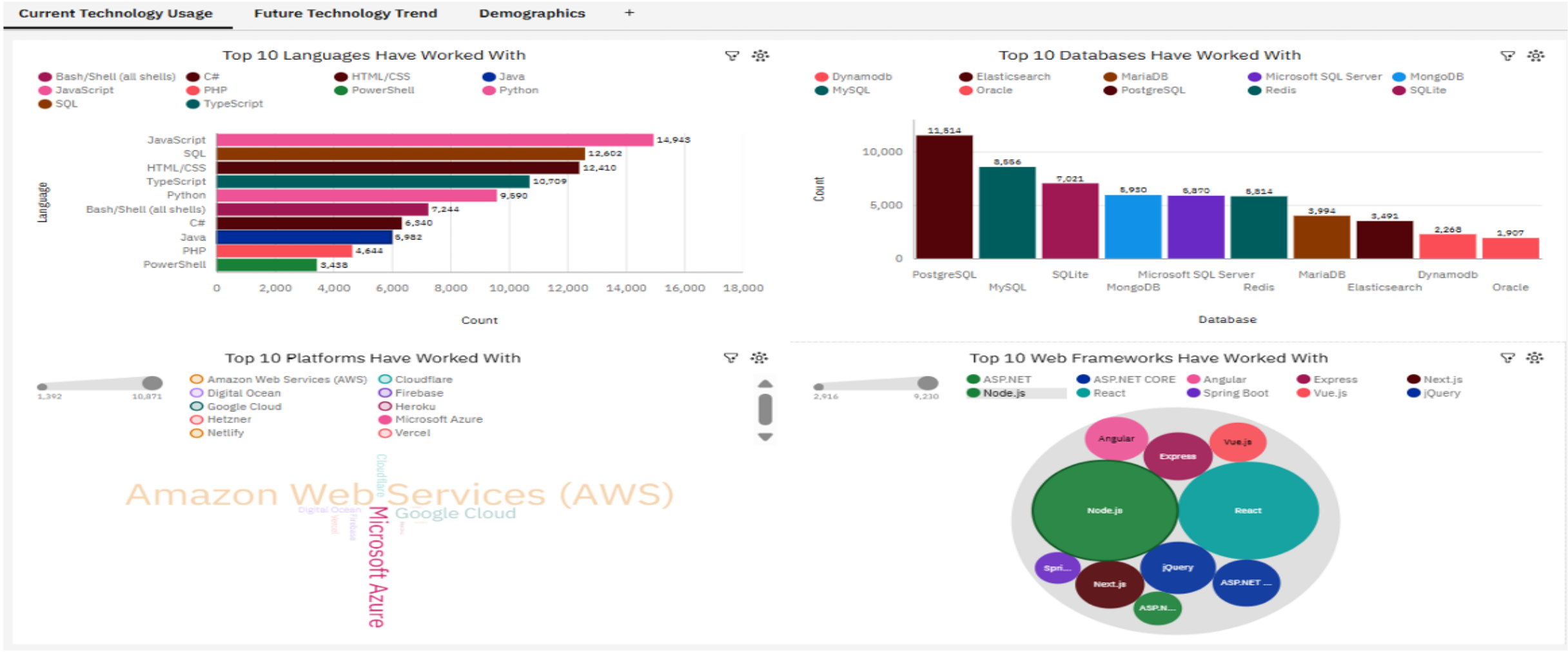
- PostgreSQL is the strongest database skill to learn or adopt
- MySQL is still widely used, but interest may be shifting toward PostgreSQL
- Redis and Supabase are worth watching as growing options

DASHBOARD



View the Survey Dashboard (PDF)

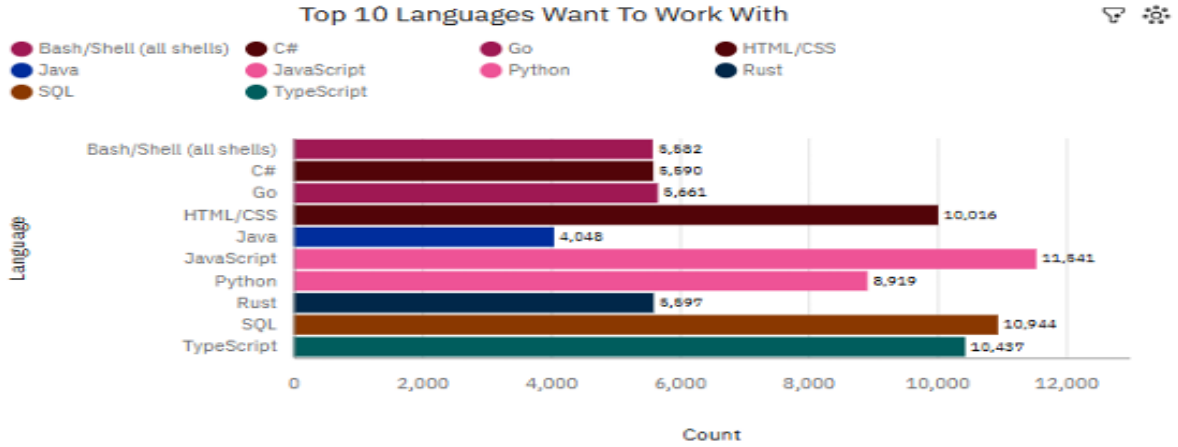
DASHBOARD TAB 1

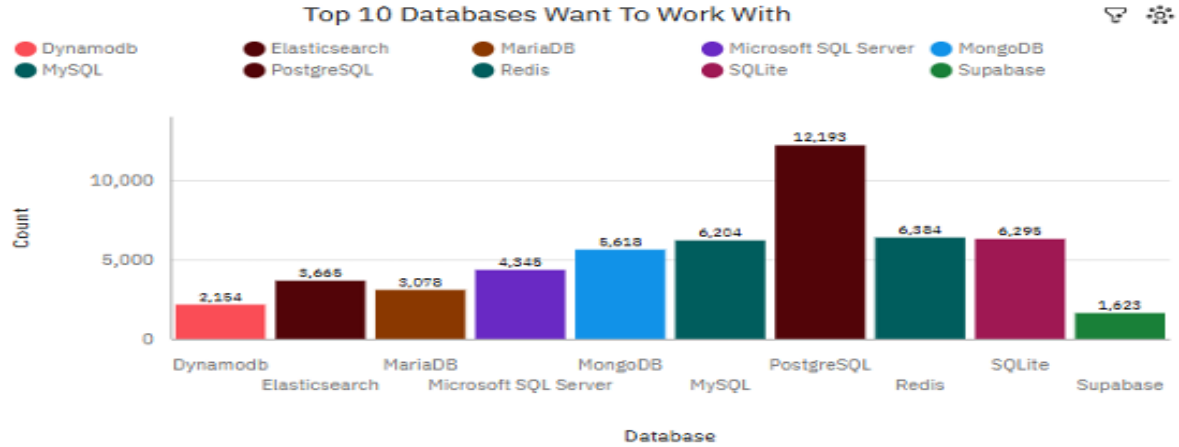


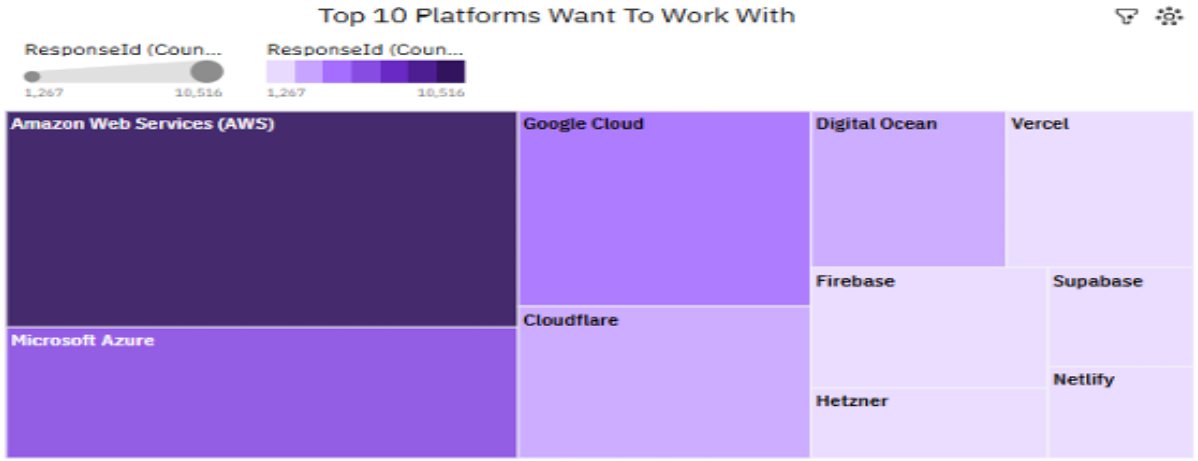
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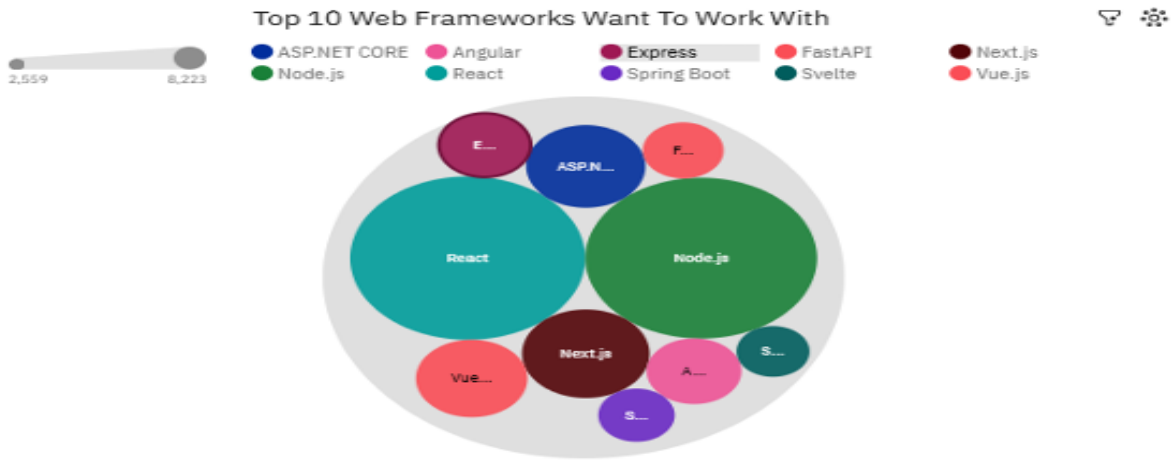
DASHBOARD TAB 2

Current Technology Usage | Future Technology Trend | Demographics | +



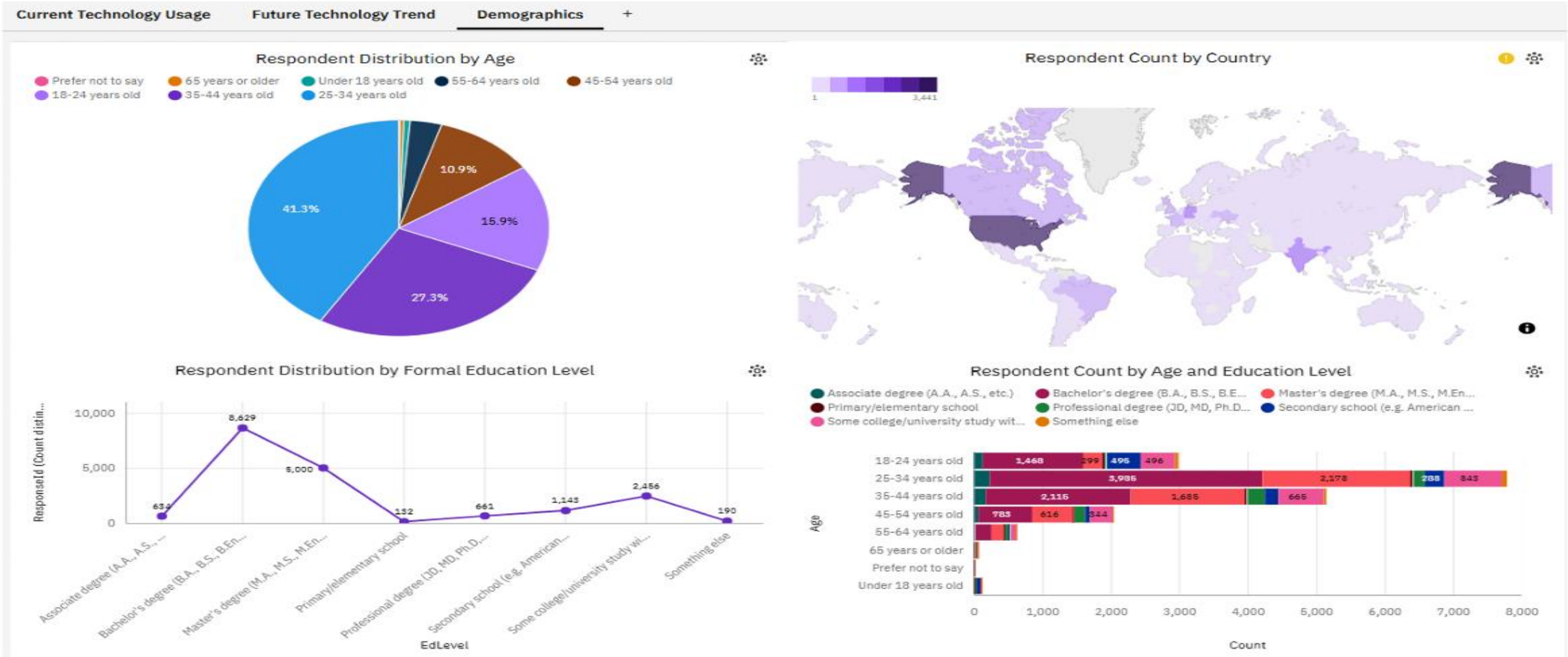






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DASHBOARD TAB 3



DISCUSSION



- **Stable core:** JavaScript, SQL and HTML/CSS lead both current use and future interest, so they are the safest skills to build on
- **Database shift:** PostgreSQL is the most wanted database (12,193), while interest in MySQL falls from 8,556 to 6,204
- **Emerging tools:** Rust, Go, Supabase and FastAPI show growing interest, though they are still smaller than the established tools
- **Cloud:** AWS leads current and future platforms, with Azure and Google Cloud next
- **Limitation:** Most respondents are aged 25-44, so the findings reflect that group and may not describe all developers

OVERALL FINDINGS & IMPLICATIONS

Findings

- JavaScript, SQL and HTML/CSS lead both current use and future interest
- PostgreSQL, AWS, Node.js and React lead their categories, today and in what developers want next
- Rust, Go, Supabase and FastAPI are rising and 25-34 year olds are the largest respondent group (41.3%)

Implications

- Web and SQL skills are the safest foundation for developers to build on
- PostgreSQL, AWS, and the Node.js/React stack are strong choices for learning or adopting
- Emerging tools are worth watching, and the findings mainly reflect younger developers

CONCLUSION



- **Core skills:** JavaScript, SQL and HTML/CSS lead both current use and future interest
- **Leading tools:** PostgreSQL, AWS, Node.js and React lead their categories today and for the future
- **Emerging trends:** Rust, Go, Supabase and FastAPI show growing interest and are worth watching
- **Next steps:** Track trends across multiple survey years and add job posting data to confirm employer demand

APPENDIX



- **Dashboard:** View the Survey Dashboard (PDF)
- **Data source:** Stack Overflow Developer Survey
- **Tools used:** SQL, Python (Jupyter Notebooks), IBM Cognos Analytics